

Performance Estimation Problems (PEPs)

Systematic analysis and design of first-order optimization algorithms
via the performance estimation framework

François Glineur, Adrien Taylor



SIAM Conference on Optimization, tutorial session – June 2026



Wooclap time!



Organization and learning outcomes (1/3)

What we would like you to learn with this tutorial:

- ◇ Base understanding of performance estimation principles (primal and dual approaches),
- ◇ leverage PEPs for numerical comparison of base first-order algorithms,
- ◇ leverage PEPs to obtain simple convergence proofs (numerics and symbolics),
- ◇ leverage PEPs for algorithm design (numerics and symbolics).

Tutorial material:

[https://github.com/PerformanceEstimation/
Tutorial-SIAM-OP26](https://github.com/PerformanceEstimation/Tutorial-SIAM-OP26)



Organization and learning outcomes (2/3)

Part 1 (now, Wed June 3, 9:15 AM - 10:45 AM):

- ◇ Talk: understand the basics of PEPs: context, problem statement, SDPs, interpolation (about 40mins)
- ◇ Hands-on: numerical comparisons of FO methods using PEPs (about 25mins)
- ◇ Talk: dual formulations and obtaining performance bounds and proofs (about 15mins)

Organization and learning outcomes (2/3)

Part 2 (this PM, Wed June 3, 3:15 PM - 4:45 PM):

- ◇ Talk: proof structures
(about 15mins)
- ◇ Hands-on: numerical and symbolic tools for proving convergence rates
(about 30mins)
- ◇ Talk: designing first-order methods: principles and an example
(about 10mins)
- ◇ Hands-on numerical and symbolic tools for designing a first-order method
(about 20mins)
- ◇ Discussions on going further
(about 15mins)

Ressources

<https://github.com/PerformanceEstimation/Tutorial-SIAM-OP26>

Tools:

- ◇ Jupyter notebooks, or colab (here).
- ◇ Optimization tools/packages:
 - convex optimization interfacing/modeling tool: CVXPY.
 - Semidefinite (SDP) solvers: get academic MOSEK if possible (here).
- ◇ Symbolic computations:
 - SymPy [**OK for this mini-tutorial**], or Wolfram Engine (here+jupyter).
 - Mathematica is better (but expensive) — free (limited) version via cloud: here.

Do interrupt with questions!